HYDRULIC TURBINE

Radial flow Reaction turbine are those turbines in which the water flows in the radial direction. The water may flow radially from outwards to inwards or from inwards to outwards.

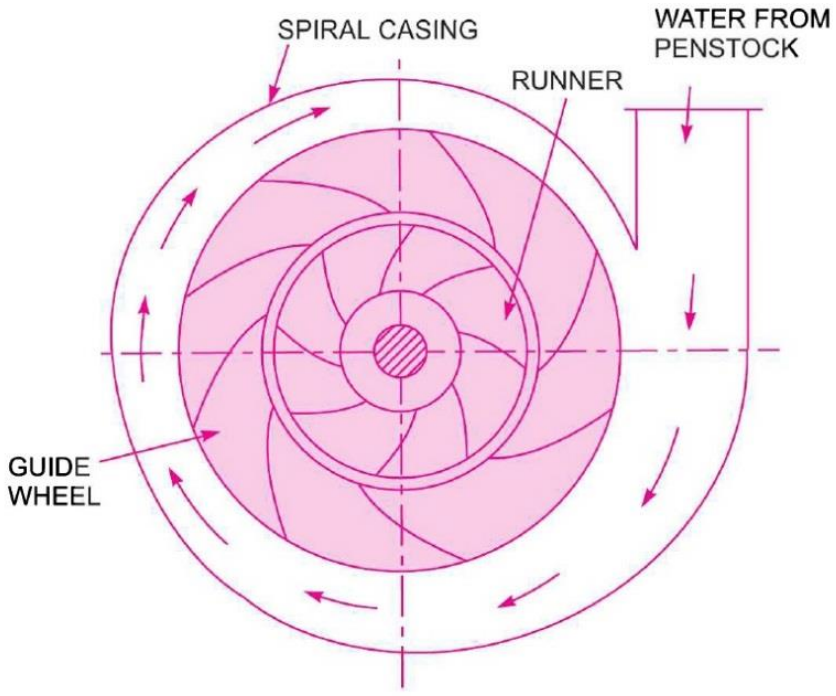
Reaction turbine means that the water at the inlet of the turbine possess kinetic energy as well as pressure energy. As the water flows through the runner, a part of pressure energy goes on changing into kinetic energy. Thus, the water through the runner is under pressure. The runner is completely enclosed in an air tight casing and casing and the runner is always full of water.

Important **reaction** turbines are **Francis**, **Kaplan** and **Propeller.**

Main parts of a Radial Flow Reaction turbine:

The main parts of a radial flow reaction turbine are:

1) Casing
2) Guide mechanism
3) Runner
4) Draft Tube

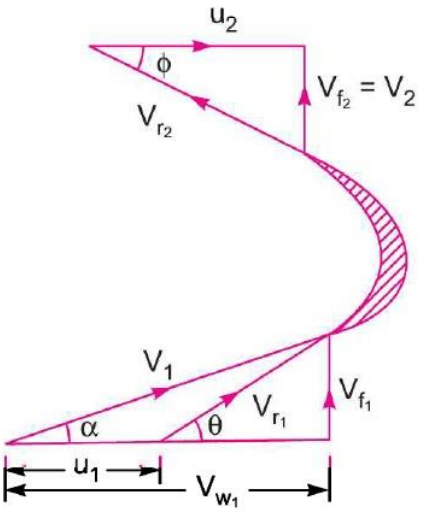


FRANCIS TURBINE:

The main parts of a Francis turbine are:

1. **Penstock:** It is a large size conduit which conveys water from the upstream of the dam/reservoir to the turbine runner.
2. **Spiral/scroll casing**: It constitutes a closed passage whose cross-sectional area gradually decreases along the flow direction. Area is maximum at inlet and nearly zero at exit.
3. **Guide Mechanism:** It consists of stationary circular wheel all round the runner of the turbine. The stationary guide vanes are fixed on the guide mechanism. It changes the position of the guide blades/vanes to affect a variation in water flow rate, when the load conditions on the turbine change.
4. **Runner:** It is a circular wheel on which a series of radial curved vanes are fixed. The surfaces of the vanes are made very smooth. The radial vanes are so shaped such that the water enters and leaves the runner without shock.
5. **Draft tube:** The pressure at the exit of the runner of a reaction turbine is generally less than atmospheric pressure. The water at the outlet cannot be directly discharged to the tail race. A tube or pipe of gradually increasing area is used for discharging water from the exit of the turbine to the tail race. This tube of increasing area is called Draft Tube.

Velocity Triangle, Work Done and Efficiency of a Francis Turbine:



In the modern Francis turbine, the water enters runner of the turbine in the radial direction at the outlet and leaves in the axial direction in the inlet of runner. Thus, the modern Francis turbines are the mixed flow type turbine.

As in the case of Francis turbine, the discharge at the outlet is radial, the velocity at the outlet i.e $\mathbf{V_{w2}}$ will be zero.

$\mathbf{V_2=V_{f2}}$

Work done:

The work done by water on the runner per second will be:

$\mathbf{=\rho Q(V_{w1}u_1+V_{w2}u_2)}$

$\mathbf{= \rho\ Q(V_{w1}u_1)}$ (since $V_{w2}=0$)

Work done per second per unit weight of water striking/s

$\mathbf{=1/g\ (V_{w1}u_1)}$

Hydraulic efficiency:

$$\eta_h = \frac{(V_{w1}u_1)}{gH}$$

Discharge at the turbine:

$\mathbf{Q=\pi D_1B_1V_{f1}= \pi D_2B_2V_{f2}}$

D_1=Diameter of the runner at inlet

B_1=Width of the runner at inlet

V_{f1}=Velocity of flow at inlet

D_2, B_2, V_{f2} are corresponding values at outlet

Points to be remember for Francis turbine:

1) The ratio of width of the wheel to its diameter is given by $n=B_1/D_1$. The value of n varies from 0.10 to 0.40
2) The flow ratio is given as,

$$\textbf{Flow ratio}=\frac{V_{f1}}{\sqrt{(2gH)}}$$

3) The speed ratio is given as

$$\textbf{Speed ratio}=\frac{u_1}{\sqrt{(2gH)}}$$